Example 4.8 Show that the set Q^+ of all positive rational numbers forms an abelian group under the operation $*$ defined by $a * b = \frac{1}{2}ab$; $a, b \in Q^+$.

When $a, b \in Q^+$, $\frac{ab}{2} \in Q^+$

$\therefore$ Q^+ is closed under the operation $*$

Now $(a * b) * c = \left(\frac{ab}{2}\right) * c = \frac{ab}{2} \cdot \frac{c}{2} = \frac{abc}{4}$

$a * (b * c) = a * \left(\frac{bc}{2}\right) = \frac{1}{2} a \cdot \frac{bc}{2} = \frac{abc}{4}$

$\therefore$ $(a * b) * c = a * (b * c)$

Hence $*$ is associative.

Let e be the identity element of Q^+ under $*$

$\therefore$ $a * e = e * a = a$, for $a \in Q^+$

i.e., $\frac{1}{2}ae = a$ i.e., $a(e - 2) = 0$

Since $a > 0$, we get $e = 2 \in Q^+$

Hence identity element exists.

Let b be the inverse of the element $a \in G$

Then $a * b = b * a = e = 2$

i.e., $\frac{1}{2}ab = 2$

$\therefore$ $b = \frac{4}{a} \in Q^+$

Thus, every element of Q^+ is invertible

$\therefore \quad (Q^+, *)$ is a group.

Also $\quad b * a = a * b \; \frac{1}{2} ab$

$\therefore \quad (Q^+, *)$ is an abelian group.

Table 11.2

Example 8. Let Z be the set of integers, show that the operation $*$ on Z, defined by $a * b = a + b + 1$ for all $a, b \in Z$ satisfies the closure property, associative law and the commutative law. Find the identity element. What is the inverse of an integer a ?

Solution. Since Z is closed for addition, as we have

$$a + b \in Z \text{ for all } a, b \in Z$$
$$\Rightarrow a + b + 1 \in Z$$
$$\Rightarrow a * b \in Z$$

So $*$ is a binary operation on Z.

Again,
$$\begin{aligned} a * b &= a + b + 1 \\ &= b + a + 1 \quad \text{(by commutative law of addition on Z)} \\ &= b * a \text{ for all } a, b \in Z \end{aligned}$$

Hence $*$ is commutative.

Again,
$$\begin{aligned} (a * b) * c &= (a + b + 1) * c \\ &= (a + b + 1) + c + 1 = (a + b + c) + 2 \end{aligned}$$

and
$$\begin{aligned} a * (b * c) &= a * (b + c + 1) \\ &= a + (b + c + 1) + 1 = (a + b + c) + 2 \end{aligned}$$

Thus
$$(a * b) * c = a * (b * c) \text{ for all } a, b, c \in Z$$

Hence, $*$ is associative.

Now, if e is the identity element in Z for $*$, then for all $a \in Z$

$$a * e = a \Rightarrow a + e + 1 = a$$
$$\Rightarrow e = -1 \in Z$$

So, -1 is the identity element for $*$ in Z.

Let the integer a have its inverse b. Then,

$$a * b = -1 \Rightarrow a + b + 1 = -1$$
$$\Rightarrow b = -(2 + a)$$

$\Rightarrow$ So, the inverse of a is $-(2 + a)$.

Lemma: *In a group G,*

(1) *Identity element is unique.*

(2) *Inverse of each $a \in G$ is unique.*

(3) $(a^{-1})^{-1} = a$, *for all $a \in G$, where a^{-1} stands for inverse of a.*

(4) $(ab)^{-1} = b^{-1}\, a^{-1}$ *for all $a, b \in G$*

(5) $ab = ac \Rightarrow b = c$

$ba = ca \Rightarrow b = c$ *for all $a, b, c \in G$*

(called the cancellation laws).

Proof: (1) Suppose e and e' are two elements of G which act as identity.

Then, since $e \in G$ and e' is identity,

$$e'e = ee' = e$$

and as $e' \in G$ and e is identity

$$e'e = ee' = e'$$

The two $\Rightarrow e = e'$

which establishes the uniqueness of identity in a group.

(2) Let $a \in G$ be any element and let a' and a'' be two inverse elements of a, then

$$aa' = a'a = e$$

$$aa'' = a''a = e$$

Now $\quad a' = a'e = a'(aa'') = (a'a)a'' = ea'' = a''.$

Showing thereby that inverse of an element is unique. We shall denote inverse of a by a^{-1}.

(3) Since a^{-1} is inverse of a

$$aa^{-1} = a^{-1}a = e62$$

which also implies a is inverse of a^{-1}

Thus $(a^{-1})^{-1} = a$.

(4) We have to prove that ab is inverse of $b^{-1}a^{-1}$ for which we show

$$(ab)\,(b^{-1}a^{-1}) = (b^{-1}a^{-1})\,(ab) = e.$$

Now
$$\begin{aligned}(ab)\,(b^{-1}a^{-1}) &= [(ab)\,b^{-1}]\,a^{-1}\\ &= [(a(bb^{-1})]\,a^{-1}\end{aligned}$$

Similarly $\quad (b^{-1}a^{-1})\,(ab) = e$

and thus the result follows.

(5) Let $ab = ac$, then

$$\begin{aligned}b &= eb = (a^{-1}a)b\\ &= a^{-1}(ab) = a^{-1}\,(ac)\\ &= (a^{-1}\,a)c = ec = c\end{aligned}$$

Thus $\quad ab = ac \Rightarrow b = c$

which is called the left cancellation law.

One can similarly, prove the right cancellation law.

Example 36. The group $(G, +_6)$ is a cyclic group where $G = \{0, 1, 2, 3, 4, 5\}$.

Solution. We see that

$1^1 = 1$, $1^2 = 1 +_6 1 = 2$, $1^3 = 1 +_6 1^2 = 3$, $1^4 = 1 +_6 1^3 = 1 +_6 3 = 4$, $1^5 = 1 +_6 1^4 = 1 +_6 4 = 5$, $1^6 = 0$

Thus $G = \{1^0, 1^2, 1^3, 1^4, 1^5, 1^6 = 0\}$

Hence G is a cyclic group and 1 is a generator.

Similarly, it can be shown that 5 is another generator.

Example 21. Let $G = \{1, -1, i, -i\}$ be a multiplicative group. Find the order of every element.

Solution. 1 is the identity element in G.

(*i*) $1^1 = 1 \Rightarrow o(1) = 1.$

(*ii*) $(-1)^2 = 1$, $(-1)^n \neq 1$ for any positive integer $n < 2$

Hence $o(-1) = 2.$

(*iii*) $(i)^4 = 1$ and $(i)^n \neq 1$ for any positive integer $n < 4.$

Hence $o(i) = 4.$

(*iv*) $(-i)^4 = 1$ and $(-i)^n \neq 1$ for any positive integer $n < 4.$

Hence $o(-i) = 4.$

Example 4.11 If $\{G, *\}$ is an abelian group, show that $(a * b)^n = a^n * b^n$, for all $a, b \in G$, where n is a positive integer.

Since, $\{G, *\}$ is an abelian group,

$$a * b = b * a \tag{1}$$

For $a, b \in G$, we have $(a * b)^1 = (b * a)^1$, by (1)

and
$$\begin{aligned}(a * b)^2 &= (a * b) * (a * b)\\ &= a * (b * a) * b, \text{ by associativity}\\ &= a * (a * b) * b, \text{ by (1)}\\ &= (a * a) * (b * b), \text{ by associativity}\\ &= a^2 * b^2\end{aligned}$$

Thus, the required result is true for $n = 1, 2$. Let us assume that the result is valid for $n = m$.

i.e.,
$$(a * b)^m = a^m * b^m \tag{2}$$

Now
$$\begin{aligned}(a * b)^{m+1} &= (a * b)^m * (a * b)\\ &= (a^m * b^m) * (a * b), \text{ by (2)}\\ &= a^m * (b^m * a) * b, \text{ by associativity}\\ &= a^m * (a * b^m) * b, \text{ since } G \text{ is abelian}\end{aligned}$$

$$= (a^m * a) * (b^m * b), \text{ by associativity}$$
$$= a^{m+1} * b^{m+1}$$

Hence, by induction, the result is true for positive integral values of n.

Example 17. Prove that if $a^2 = a$, then $a = e$, a being an element of a group.

Solution. Let a be an element of a group G such that $a^2 = a$.

To prove that $a = e$.

$$a^2 = a \Rightarrow a.a = a \Rightarrow (aa)\, a^{-1} = aa^{-1}$$

$$\Rightarrow a\,(aa^{-1}) = e. \qquad (\because\ aa^{-1} = e)$$

$$\Rightarrow ae = e \Rightarrow a = e. \qquad (\text{since } ae = a.)$$

Example 18. Show that if every element of a group (G, o) be its own inverse, then it is an abelian group.

Is the converse true ?

Solution. Let $a, b \in$ G, then $a \circ b \in$ G (closure).

Hence, by the given condition, we have

$$\begin{aligned} a \circ b &= (a \circ b)^{-1} \\ &= b^{-1} \circ a^{-1} \\ &= b \circ a, \text{ since } a^{-1} = a \text{ and } b^{-1} = b. \end{aligned}$$

Thus $a \circ b = b \circ a$, for every $a, b \in$ G.

Therefore, it is an abelian group.

The converse is not true. For example, (R, +), where R is the set of all real numbers, is an abelian group, but no element except 0 is its own inverse.

Example 19. Show that if a, b are arbitrary elements of a group G, then $(ab)^2 = a^2b^2$ if and only if G is abelian.

Solution. Let a and b be arbitrary elements of a group G. Suppose $(ab)^2 = a^2 b^2$. ... (1)

To prove G is abelian, we have to show that

$$ab = ba$$

$$(ab)^2 = a^2b^2 \Rightarrow (ab)\,(ab) = (aa)\,(bb)$$

$$\Rightarrow a\,(ba)\,b = a\,(ab)\,b, \quad \text{by associative law}$$

$$\Rightarrow (ba)\,b = (ab)\,b, \quad \text{by left cancellation law}$$

$$\Rightarrow ba = ab, \quad \text{by right cancellation law.}$$

Again, suppose G is abelian so that

$$ab = ba \quad \forall\ a, b \in G \qquad \text{.... (2)}$$

To prove that $(ab)^2 = a^2b^2$

$$(ab)^2 = (ab)\,(ab) = a\,(ba)\,b = a\,(ab)\,b, \qquad \textit{by } (2)$$

$$= (aa)\,(bb) = a^2b^2.$$

Hence proved.

Example 14. Let $G = \{(a, b) \mid a, b \in R, a \neq 0\}$. Define a binary operation $*$ on G by

$$(a, b) * (c, d) = (ac, bc + d)$$

for all $(a, b), (c, d) \in G$. Show that $(G, *)$ is a group.

Closure Property : Let (a, b) and (c, d) be any two members of G. Then $a \neq 0$ and $c \neq 0$. Therefore, $ac \neq 0$. Consequently $(a, b) * (c, d) = (ac, bc + d)$ is also a member of G. Hence G is closed with respect to the given composition.

Associative law : Let (a, b), (c, d) and (e, f) be any three members of S. Then

$$\begin{aligned} [(a, b) * (c, d)] * (e, f) &= (ac, bc + d) * (e, f) \\ &= ([ac]\, e, [bc + d]\, e + f) \\ &= (ace, bce + de + f). \end{aligned}$$

Also $$\begin{aligned} (a, b) * [(c, d) * (e, f)] &= (a, b) \times (ce, de + f) \\ &= (a\,[ce], b\,[ce] + de + f) \\ &= (ace, bce + de + f). \end{aligned}$$

Hence the given composition * is associative.

Identity element : Suppose (x, y) is an element of G such that $(x, y) * (a, b) = (a, b)$ $\forall$ $(a, b) \in G$

Then $(xa, ya + b) = (a, b)$. Hence $xa = a$ and $ya + b = b$.

These give $x = 1$ and $y = 0$. Now $(1, 0) \in G$

Therefore, $(1, 0)$ is the identity element

Inverse element : Let (a, b) be any member of G. Let (x, y) be a member of G such that $(x, y) \times (a, b) = (1, 0)$, then (x, y) be the inverse of (a, b).

Then $(xa, ya + b) = (1, 0)$. Hence $xa = 1$, $ya + b = 0$.

These give $x = 1/a$, $y = -\, b/a$.

Since $a \neq 0$, therefore, x and y are real numbers.

Also $x = \dfrac{1}{a} \neq 0$. Thus $\left(\dfrac{1}{a}, -\dfrac{b}{a}\right)$ is the inverse of (a, b).

Hence G is a group.

Note. In the above group, we have

$$(a, b) * (c, d) = (ac, bc + d)$$

and $$(c, d) * (a, b) = (ca, da + b).$$

Thus, in general, $(a, b) * (c, d) \neq (c, d) * (a, b)$ *i.e.*, the composition is not commutative and hence the group is not abelian.

Example 10: Let $G = \{(a, b) \mid a, b \text{ rationals}, a \neq 0\}$. Define $*$ on G by

$$(a, b) * (c, d) = (ac, ad + b)$$

Closure follows as $a, c \neq 0 \Rightarrow ac \neq 0$

$$\begin{aligned}[(a, b) * (c, d)] * (e, f) &= (ac, ad + b) * (e, f)\\ &= (ace, acf + ad + b)\\ (a, b) * [(c, d) * (e, f)] &= (a, b) * (ce, cf + d)\\ &= (ace, acf + ad + b)\end{aligned}$$

proves associativity.

$(1, 0)$ will be identity and $(1/a, -b/a)$ will be inverse of any element (a, b).

G is not abelian as

$$(1, 2) * (3, 4) = (3, 4 + 2) = (3, 6)$$

$$(3, 4) * (1, 2) = (3, 6 + 4) = (3, 10).$$

Problem 13: *Show that union of two subgroups may not be a subgroup.*

Solution: Let $H_2 = \{2n \mid n \in \mathbf{Z}\}$

$$H_3 = \{3n \mid n \in \mathbf{Z}\}$$

where $(\mathbf{Z}, +)$ is the group of integers. H_2 and H_3 will be subgroups of $\mathbf{Z}$. Indeed

$$2n - 2m = 2(n - m) \in H_2$$

Now $H_2 \cup H_3$ is not a subgroup as $2, 3 \in H_2 \cup H_3$

but $$2 - 3 = -1 \notin H_2 \cup H_3$$

Can union of two subgroups be a subgroup? The answer is provided by

Theorem 12: *Union of two subgroups is a subgroup iff one of them is contained in the other.*

Theorem 12.18. Every cyclic group is an abelian group.

Solution. Let G be a cyclic group and let a be a generator of G so that

$$G = \langle a \rangle = \{a^n : n \in Z\}$$

If g_1 and g_2 are any two elements of G, there exist integers r and s such that $g_1 = a^r$ and $g_2 = a^s$. Then

$$g_1 g_2 = a^r a^s = a^{r+s} = a^{s+r} = a^s \,.\, a^r = g_2 g_1$$

So, G is abelian.

3. If $\{G, *\}$ is a finite cyclic group generated by an element $a \in G$ and is of order n, then $a^n = e$ so that $G = \{a, a^2, \ldots, a^n(= e)\}$. Also n is the least positive integer for which $a^n = e$.

Proof

If possible let there exist a positive integer $m < n$ such that $a^m = e$.

Since G is cyclic, any element of G can be expressed as a^k, for some $k \in Z$.

When k is divided by m, let q be the quotient and r be the remainder, where $0 \leq r < m$.

Then $$k = mq + r$$

$\therefore$ $$\begin{aligned} a^k = a^{mq+r} &= a^{mq} * a^r \\ &= (a^m)^q * a^r \\ &= e^q * a^r \\ &= e * a^r \\ &= a^r \end{aligned}$$

This means that every element of G can be expressed as a^r, where $0 \leq r < m$.

This implies that G has at most m elements or order of $G = m < n$, which is a contradiction.

i.e., $$a^m = e, \text{ for } m < n \text{ is not possible.}$$

Hence $a^n = e$, where n is the least positive integer. Now let us prove that the elements $a, a^2, a^3, \ldots, a^n$ $(= e)$ are distinct.

If it is not true, let $a^i = a^j$, for $i < j \leq n$

Then $$a^{-i} * a^i = a^{-i} * a^j$$

i.e., $$e = a^{j-i}, \text{ where } j - i < n,$$

which again is a contradiction.

Hence $$a^i \neq a^j, \text{ for } i < j \leq n.$$

Theorem 18: *Order of a cyclic group is equal to the order of its generator.*

Proof: Let $G = \langle a \rangle$ *i.e.*, G is a cyclic group generated by a.

Case (*i*): $o(a)$ is finite, say n, then n is the least +ve integer s.t., $a^n = e$.

Consider the elements $a^0 = e, a, a^2, ..., a^{n-1}$

These are all elements of G and are n in number.

Suppose any two of the above elements are equal

say $\quad a^i = a^j$ with $i > j$

then $\quad a^i \,.\, a^{-j} = e \Rightarrow a^{i-j} = e$

But $0 < i - j \leq n - 1 < n$, thus $\exists$ a +ve integer $i - j$, s.t., $a^{i-j} = e$ and $i - j < n$, which is a contradiction to the fact that $o(a) = n$.

Thus no two of the above n elements can be equal, *i.e.*, G contains at least n elements. We show it does not contain any other element. Let $x \in G$ be any element. Since G is cyclic, generated by a, x will be some power of a.

Let $\quad x = a^m$

By division algorithm, we can write

$$m = nq + r \quad \text{where } 0 \leq r < n$$

Now $\quad a^m = a^{nq+r} = (a^n)^q \,.\, a^r = e^q .\, a^r = a^r$

$$\Rightarrow x = a^r \qquad \text{where } 0 \leq r < n$$

i.e., x is one of $a^0 = e, a, a^2, ..., a^{n-1}$

or G contains precisely n elements

$$\Rightarrow o(G) = n = o(a)$$

Case (*ii*): $o(a)$ is infinite.

In this case no two powers of a can be equal as if $a^n = a^m$ $(n > m)$ then $a^{n-m} = e$, *i.e.*, it is possible to find *a* +*ve* integer $n - m$ s.t., $a^{n-m} = e$ meaning thereby that a has finite order.

Hence no two powers of a can be equal. In other words G would contain infinite number of elements.

Theorem 12.19. If a is a generator of a cyclic group G, then a^{-1} is also a generator of G.

Proof. Let G = < a > be a cyclic group generated by a. Let a^r be any element of G, where r is some integer. We can write $a^r = (a^{-1})^{-r}$. Since $-r$ is also some integer, therefore each element of G, is generated by a^{-1}. Thus a^{-1} is also a generator of G.

Example 60. If a group G has four elements, show that it is Abelian.

Solution. Let $G = \{e, a, b, c\}$ be a multiplicative group of order four where e is the identity element. Since $e^{-1} = e$, there must be at least one more element in G which is its own inverse.

Let $a^{-1} = a$. If $b^{-1} = b$ and $c^{-1} = c$ then G is Abelian

If $b^{-1} = c$, then $c^{-1} = b$ and we have $bc = cb = e$. Now $a^{-1} = a \Rightarrow aa = e$.

Observe that $ab = a \Rightarrow b = e$ and $ab = b \Rightarrow a = e$ which are not possible. So, $ab = c$. Similarly, $ba = c$, $ac = ca = b$, $bb = a$ and $cc = b$.

The composition table for G is a follows:

	e	a	b	c
e	e	a	b	c
a	a	e	c	b
b	b	c	a	e
c	c	b	e	b

From this composition table it is clear that G is commutative, *i.e.*, G is an Abelian group.

Example 34. The set of integers with respect to + *i.e.,* $(Z, +)$ is a cyclic group, a generator being 1.

Solution. We have $1^0 = 1$, $1^1 = 1$, $1^2 = 1 + 1 = 2$, $1^3 = 1 + 1 + 1 = 3$ and so on.

Similarly 1^{-1} = inverse of $1 = -1$

$$1^{-2} = (1^2)^{-1} = -2,\ 1^{-3} = (1^3)^{-1} = (3)^{-1} = -3 \text{ and so on.}$$

Thus each element of G can be expressed as some integral power of 1.

Similarly we can show that -1 is also a generator.

Example 37. How many generators are there of the cyclic group G of order 8?

Solution. Let a be generator of G. Then o $(a) = 8$. We can write G = $\{a, a^2, a^3, a^4, a^5, a^6, a^7, a^8\}$

7 is prime to 8, therefore, a^7 is also a generator of G.

5 is prime to 8, therefore, a^5 is also a generator of G.

3 is prime to 8, therefore, a^3 is also a generator of G.

Thus there are only four generators of G *i.e.*, a, a^3, a^5, a^7

Example 38. Show that the group $(\{1, 2, 3, 4, 5, 6\}, \times_7)$ is cyclic. How many generators.

Solution. G be a given group. If there exists an element $a \in$ G such that $o(a) = 6$ *i.e.*, equal to the order of the group G then the group G will be a cyclic group and a will be a generator of G.

Note that $o(3) = 6$ because $3^1 = 3$, $3^2 = 3 \times_7 3 = 2$, $3^3 = 3^2 \times_7 3 = 6$, $3^4 = 6 \times_7 3 = 4$, $3^5 = 4 \times_7 3 = 5$, $3^6 \times_7 3 = 5 \times_7 3 = 1$ *i.e.*, the identity element.

So, G is cyclic and 3 is a generator of G. We can write

$$G = \{3, 3^2, 3^3, 3^4, 3^5, 3^6\}.$$

Now 5 is prime to 6. There 3^5 *i.e.*, 5 is also generator of G.